PURPOSE

The purpose of this procedure is to provide safety guidelines for masonry and concrete construction, and to protect employees and property from harm when these are conducted.

The objective of this procedure is to prevent incidents related to masonry and concrete construction performed by employees of **Richco International Inc.**

Cutting, Coring and Drilling operations will adhere to the Hot Work Program

SITE PREPARATION

Before erecting formwork or shoring the surface area in which the formwork or shoring will be placed will be prepared to receive the load that will be imposed on its surface. The existing ground will be level and thoroughly compacted before erecting formwork and shoring to prevent settlement.

The site layout will simplify the erection and taking apart of formwork and shoring, moving equipment (including ready-mix trucks), concrete pumps, canes, and the storage of reinforcement steel, formwork, and shoring.

Electrical hazards (overhead power lines and temporary power) in the work area will be identified and the local power company consulted to set up safe clearance distances or to move the utilities.

Whenever possible, formwork and shoring will be roped off from other work (such as excavation or pile driving).

REINFORCING STEEL

Reinforcing steel for walls, piers, columns, and similar structures will be laterally supported to resist overturning. The lateral supports for reinforcing steel will be capable of withstanding the force that will be applied to them during construction.

Bundles of reinforcing steel moved by crane will be securely tied together to prevent slipping. Steel over 20 feet long will be handled by two-part slings.

Exposed rebar, onto or into which workers could fall, will be covered to remove the hazard of impalement. Reinforced plastic cap coverings will be during construction.

Bundles of reinforcing steel moved by crane will be securely tied together to prevent slipping. Steel over 20 feet long will be handled by two-part slings.

Exposed rebar, onto or into which workers could fall, will be covered to remove the hazard of impalement. Reinforced plastic cap coverings will be used for impalement protection. Wooden troughs or other substantial material may also be used to cover any vertically protruding rebar.

When working more than 6 feet above any close working surfaces, placing and tying reinforcing steel in walls, piers, columns, etc., workers must use a personal fall arrest system as set forth in Company safety procedure (Fall Prevention & Protection). Position devices for rebar work will be rigged so a worker cannot free fall more than 2 feet. The positioning device will be secured to an anchorage capable of supporting at least twice the potential impact load of the worker's fall or 3,000 pounds, whichever is greater.

Reinforcing mats used as walkways will be provided with planking to provide safe footing.

Reinforcing steel will not be used as guy attachments at dead men or other anchorage points and will not be used for scaffolding hooks, stirrups, or as a load-bearing member of any lifting device.

Wire mesh reinforcing mats will be secured at each end or the roll turned over to prevent recoiling action. Unrolled wire mesh will be secured on each side of a proposed cut before cutting the mesh.

No workers, except those essential to post-tensioning operations, will be allowed to be behind the jack during tensioning. Signs and barricades will be erected to limit workers access to post-tensioning areas during tensioning.

VERTICAL SHORING

Before erection, shoring equipment will be inspected by the concrete contractor to verify that it conforms to the equipment named in the shoring layout. Unauthorized changes or substitution of equipment will not be made unless the designer has approved the change or substitution.

The manufacturer's specification for fabricated shoring will be available at the job site during the planning and execution.

Erecting shoring will be under the supervision of an experienced and competent person.

Shoring equipment will be inspected following the manufacturer's procedures. Metal frame shoring equipment and accessories will not be used if excessively rusted, bent, dented, re-welded beyond the original factory weld locations, or if they have broken welds or other defects. Each part will be in good working order and in a condition similar to that of original manufacture. Damaged shoring equipment will not be used for shoring. Any part which cannot be brought into

proper alignment or contact with the part, into or onto which it is intended to fit, will be "Red-Tagged", removed and replaced.

Erected shoring equipment will be inspected by the contractor immediately before, during, and immediately after the placement of concrete to decide the shoring equipment meets the needs named on the formwork drawings. Acceptable shoring will be "Green-Tagged". Any shoring found to be damaged or weakened must be immediately reinforced or re-shored.

Re-shoring will be provided when necessary to safely support slabs and beams after stripping or whenever the concrete is required to support loads in excess of it capacity.

Temporary storage of reinforcing rods, materials, or equipment on top of formwork is banned unless these temporary structures have been designed or strengthened to support the added loading. Eccentric loads on shore heads and similar members will be restricted unless these members are designed for such loads.

FRAME SHORING

Frame shoring (tubular welding and tube and coupler) will not be loaded beyond the safe working load recommended by the manufacturer. Frame shoring will be designed with a minimum safety factor of 2.5.

Locking devices on frames and braces will be in good working order; coupling pins will align the frame or panel legs; pivoted cross braces will have their center pivot in place and parts will be in good serviceable condition. Couplers (tube and couple shoring) will be of a type of metal such as drop-forged steel, malleable iron, or structural grade aluminum; gray cast iron will not be used. Couplers that are deformed, broken, or have defective or missing threads or bolts will not be used. Frames and braces found to be defective or damaged will be "Red-Tagged" and removed from service.

Following erection, a thorough inspection will be made to verify that:

- The shoring has been erected as shown on the layout drawing
- Spacing between towers and cross brace spacing does not exceed that shown on the layout, and that all-locking devices are in the closed position
- The devices for attaching the external lateral stability bracing are securely fasten to the legs of the shoring frame
- Interlocking tubular members and coupling are properly installed and tightened
- Base plates, shore heads, extension devices, or adjustment screws are in firm contact with the footing sill and the form
- Acceptable shoring will be "Green-Tagged"

The following general safety precautions will apply to frame shoring:

- Follow the shoring layout drawing and do not omit needed parts
- Do not exceed the shore frame spacing or tower heights as shown on the shoring layout
- The shoring load must be carried on all legs
- Plumb and level shoring frames as the erection continues and check plumb and level of shoring towers just before pour.
- Do not force braces on frames to fit level the shoring towers until proper fit can be made easily
- Tie high towers of shoring frames together with sufficient braces to make a rigid, solid unit. Shoring must always be secured when the height of the shoring towers exceeds four times the minimum base width. California (and other states) requires a height-to-minimum base width ratio of three to one (3:1). Refer to the governing codes for your job location
- Exercise caution in erecting or taking apart freestanding shoring towers to prevent tipping
- Do not climb cross braces
- Use screw jacks to adjust for uneven grade conditions, to level and accurately position the falsework, and to aid stripping
- Do not exceed the manufacturer's recommended maximum extension of screw jacks. Keep screw jack extensions to a minimum for maximum load carrying capacity
- Make certain that screw jacks are firmly in contact with the foundation and frame legs
- Screw jacks will not be used to raise formwork during concrete placement

SINGLE-POST SHORES

Single-post shoring layout will provide for the maximum intended loading with a minimum safety factor of three. When single-post shores are to be used in more than one tier, they will be designed and inspected by a registered structural engineer.

Single-post shores will be horizontally braced in both the longitudinal and transverse directions and will also be braced diagonally. The bracing will be installed as the shores are being erected.

Single-post shoring layouts will be horizontally braced in both the longitudinal and transverse directions and will also be braced diagonally. The bracing will be installed as the shores are being erected.

Single-post shores and adjusting devices will be inspected before use. Fabricated shores and adjusting devices will not be used if heavily rusted, bent, dented, re-welded, damaged, or deficient in any manner. Timber shores and timber components of fabricated shores will not be used if split, knotted, broken, or otherwise structurally deficient.

Base plates and shore heads of single-post shores will be in firm contact with the footing sill and the form material.

Adjustment of single-post shores will not be made after the concrete is in place.

RELEASING AND MOVING FORMS

Forms will be securely attached to wire rope slings, having a minimum safety factor of eight, when raised or moved by crane or other mechanical lifting devices. Panels and form sections will be equipped with hoisting brackets or attachments for slings.

Loose tools and materials will be removed before moving the forms. Taglines for controlling forms will be used whenever necessary to protect workers or structures.

Workers are not allowed to ride forms or form scaffolding being raised or moved.

Vertical and overhead forms will not be released until adequately braced or secured. Workers at lower levels exposed to falling materials will be removed to a safe area before release or moving forms.

Forms, shores and bracing (except those used for slab on grade and slipforms) will not be removed until the concrete has gained enough strength as listed in the contract specifications or shown on the form drawings.

CONCRETE TOOLS and CONVEYANCE SYSTEMS

Concrete buckets will have positive safety latches or similar safety devices installed to prevent premature or accidental dumping. Manually operated gates will be of the self-closing type. Riding the concrete bucket is restricted. Raised concrete buckets will be routed so no worker, or the fewest number possible, are exposed to the hazard of falling concrete buckets. An appointed signal person will provide direction to the crane operator using the standard hand signals or radio communications that are common to the industry. Cranes and rigging must comply with the Company policies outlined in Company safety procedure (Cranes) as well as the regulations of the Occupational Safety and Health Administration (OSHA).

Handles on bull floats used where they can contact energized electrical conductors must be made of nonconductive material or insulated with a nonconductive sheath that will protect the operation from electrical shock.

Powered and rotating type concrete troweling machines that are manually guided must be equipped with a control switch that has a positive mechanical release (dead-man switch) that automatically stops trowel rotation when the operator removes his and her hand from the equipment handle.

Handles of concrete buggies must not extend beyond the wheels on either side of the buggy. Motorized concrete buggies will be equipped with a dead-man control switch and knuckle guards. Stop-checks will be used at all places where buggies dump.

Runways will be of sturdy construction, evenly supported and will have a smooth running surface and curbs (2" x 2" or 4" x 4"). Where necessary, runways will have a railing high enough on the open side to protect workers. If a single runway is used, turnarounds will be provided. Buggies will be routed in a continuous loop to lessen the danger of collision.

Sections of trimies and similar concrete conveyances will be secured with wire rope or equivalent material as well as the regular coupling. Cup-type couplings using flanges with the same outer diameter as pipe are restricted.

Concrete trucks must not travel through the job site with the chute extended. When unloading on a slope, the wheels of the concrete truck will be blocked, and the brakes set to prevent movement.

Concrete mixers with one cubic yard or larger loading skips will be equipped with a mechanical device to clear the skip of material. Skip clearing will not be performed by a worker standing under or near a raised skip while striking it with a handheld tool. Guardrails will be provided on each side of the dangerous area under the raised skip

Grid-guards will be installed on all motor, plaster or fireproof mixers of one yard capacity or smaller. Any guards will be in place before the mixer is ran. The mixer will be locked out when work (i.e., clean out) is performed on the mixer.

Bulk cement storage structures will be equipped with conical or tapered bottoms, and mechanical or pneumatic means of starting the flow of material. No worker will be allowed to enter storage facilities unless the ejection system has been shut down, locked out and tagged to suggest the ejection system is not to be worked.

Masonry saws will be guarded with a semicircular enclosure over the blade.

No worker will be allowed to perform maintenance or repair activities on equipment (i.e., mixers, screens or pumps used for concrete and masonry construction) where the unplanned operation of the equipment could occur and cause injury, unless all known potentially hazardous energy sources have been locked out and tagged.

CONCRETE PUMP TRUCKS

Concrete pump trucks and parts (pump, boom, piping) will be inspected by the manufacturer's instructions and found to be in safe working condition. Concrete pumps will be equipped with an emergency shut off.

Trucks will be positioned so any obstacles (power lines, ditches, walls, and columns) do not interfere with safe operation. Whenever possible, at least 17 feet clearance will be kept from power lines. Never less than 10 feet of clearance will ever occur.

Visual communication between the pump operator and the placement area will be kept, or a signalperson will be used to assume safe placement of concrete.

Outriggers must be extended and locked into place and kept away from un-supportive surfaces. Pads or dunnage will be used as necessary to stabilize the crane. Raised tires will be blocked to prevent unnecessary bouncing or rolling.

The boom must never be allowed to contact deck forming or near structures.

Clamps, pipe, safety straps, and restraining devices on hoses suspended from booms will be inspected. Cup-type couplings using flanges with same outer diameter as pipe are restricted.

Concrete pumps will not be worked unless the hopper is guarded. Workers will not place hands or objects in the hopper unless it is locked out.

Pumping line clean out operation must conform to the manufacturer's needs and will be conducted in a named area. The use of the compressed air procedure for cleaning pumping lines is restricted.

MASONRY CONSTRUCTION

A limited-access zone will be set up whenever a masonry wall is being built. The limited-access zone will be set up before building the wall and will be equal to the height of the wall being built, plus four feet, and will run the entire length of the wall.

The limited-access zone will be set up on the side of the wall opposite the scaffolding.

Employees who are building the wall they may enter the limited-access zone, but will not

allow
other employees to enter. The limited-access zone will remain in place until the wall is adequately supported to prevent overturning or collapse.

Masonry walls over eight feet in height will be adequately braced and supported so they will not overturn or collapse. Limited-access zones and bracing will remain in intact until permanent supporting elements of the structure are in place.

Scaffolding for masonry construction will meet or exceed the needs of 29 CFR 1926 Subpart L and Company safety procedure (Scaffolds).

HEALTH HAZARD

The cutting, grinding, drilling and finishing of concrete and masonry poses a potential silica dust hazard. Each employer must warn employees about the hazards posed by silica dust. Where concrete tools (jackhammers, masonry saws, grinders, etc.) disturb masonry and concrete products, engineering or work practice controls must be implemented to reduce (silica) dust.

When engineering or work practice controls are not possible, employees exposed to dust will use respiratory protection. Suggested engineering controls and work practice controls for preventing silica dust is as follows:

- Post Warning Signs
- Employ wet methods
- Connect the tool to a point-of-operation dust collection system
- Limit the number of workers in the work area (workers who are in the area must use respiratory protection)
- Contain the work area with barricades and similar devices
- Use a ventilation system that removes and collects dust (workers in containment must use respiratory protection). Point exhaust away from other workers

Concrete and masonry cement is irritating to the eyes and skin. Protective equipment such as gloves, rubber boots, kneepads, and protective eyewear must be used when handling concrete. Washing with soap and water is important to prevent skin irritation. Hands will not be washed in a bucket of water used for moistening and washing concrete tools.

Safety data sheets (SDS) will be available for concrete and masonry products that will be used or disturbed. Also, SDSs are needed for form-release agents, concrete additives, or cure agents. All containers will be labeled.

TRAINING

Employees working with or near concrete and masonry operations may not work until the following training has been completed and documented.

- Concrete Operation and Safety Procedures
- Proper tool use, inspection, and maintenance
- Health Hazards associated with concrete and masonry
- Required PPE and clothing
- Sanitation and Decontamination Procedures
- Hazard Communication
- Fall Protection
- Vehicle Safety
- Access Zones and Traffic Control